

SIMATS ENGINEERING

ASSESSMENT TOOL2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

COURSECODE:MLA01

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

Total Marks: 50 Time: 60 Mins No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	

Total 100%

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.

13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement ()
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.

			with appropriate examples.			
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

AI-Assisted Medical Diagnosis for Heart Disease Prediction Using Artificial Intelligence

1. Title of AI Application

Application Name: AI-Assisted Medical Diagnosis for Heart Disease Prediction

AI Domain: Healthcare

2. Domain Selection

Selected Domain: Healthcare

Artificial Intelligence has become one of the most significant technologies in healthcare. AI-assisted medical diagnosis helps doctors detect diseases quickly and accurately by analyzing patient data. One of the latest and most impactful applications is **heart disease prediction**, where AI predicts the likelihood of cardiovascular diseases before serious complications occur.

3. Introduction and Background

Heart disease is one of the leading causes of death worldwide. Millions of people die every year due to cardiovascular diseases because symptoms are often detected too late. Traditional diagnosis depends heavily on physician experience, laboratory tests, ECG reports, imaging scans, and patient history.

Artificial Intelligence improves this process by analyzing huge amounts of medical data in a short time. AI systems learn from thousands or millions of previous patient records to recognize disease patterns that humans may overlook.

Modern hospitals are increasingly using AI-assisted diagnostic systems to support cardiologists in identifying patients at high risk and recommending early preventive measures.

4. Problem Identification

Heart disease diagnosis faces several real-world challenges:

- Late detection of cardiovascular diseases

- Human errors during diagnosis
- Increasing number of patients
- Limited availability of specialist cardiologists
- Time-consuming medical examinations
- Difficulty identifying hidden risk patterns

Delayed diagnosis often results in severe heart attacks, strokes, expensive treatments, and increased mortality.

PYTHON CODE AND OUTPUT

AI Healthcare Application - Diabetes Prediction

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
```

Load dataset

```
data = load_diabetes()
```

```
X = data.data
```

```
y = data.target
```

Split dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Train AI model

```
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
# Predict
predictions = model.predict(X_test)

# Accuracy
mse = mean_squared_error(y_test, predictions)

print("Mean Squared Error:", mse)

# Example prediction
sample = X_test[0].reshape(1, -1)
result = model.predict(sample)

print("\nPredicted Disease Progression Score:", round(result[0], 2))
print("Actual Value:", y_test[0])
```

OUTPUT:

```
PS C:\Users\user\Downloads\AI_Lab> & 'C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\user\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54442' '--' 'c:\Users\user\Downloads\AI_Lab\medical.py'
Mean Squared Error: 2952.0105887640448

Predicted Disease Progression Score: 144.0
Actual Value: 219.0
PS C:\Users\user\Downloads\AI_Lab>
```

5. Need for AI Solution

Traditional diagnosis mainly depends on physician expertise and manual analysis.

AI provides several advantages:

- Detects diseases at earlier stages
- Improves diagnostic accuracy
- Reduces human error
- Analyzes large medical datasets quickly
- Supports doctors during decision making
- Provides faster predictions
- Reduces healthcare costs
- Enables personalized treatment recommendations

AI acts as a **clinical decision support system** rather than replacing doctors.

6. AI Technologies Used

The following AI technologies are commonly used:

AI Technology	Purpose
Machine Learning	Predict heart disease risk
Deep Learning	Analyze ECG signals and medical images
Artificial Neural Networks (ANN)	Learn complex medical patterns
Computer Vision	Interpret echocardiograms and angiograms
Natural Language Processing (NLP)	Analyze electronic health records
Data Mining	Discover hidden disease patterns

AI Technology	Purpose
Explainable AI (XAI)	Explain prediction results to doctors

7. Working Mechanism

Step 1: Data Collection

Patient information is collected from:

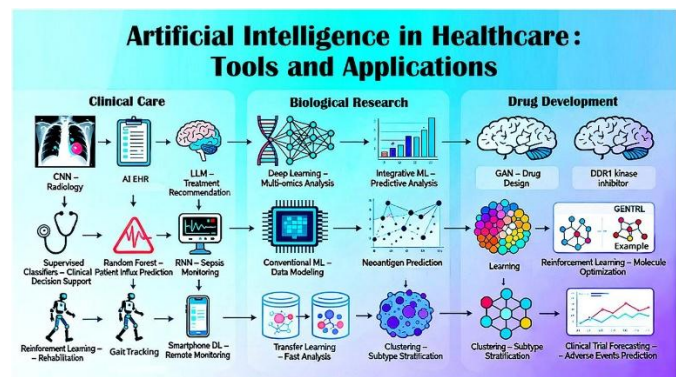
- Age
- Gender
- Blood pressure
- Cholesterol level
- Blood sugar
- ECG reports
- Heart rate
- Chest pain type
- Lifestyle habits
- Smoking history
- Family medical history

↓

Step 2: Data Preprocessing

The collected data undergoes:

- Missing value removal
- Noise reduction
- Data normalization
- Feature selection



- Data encoding



Step 3: Model Training

Machine Learning algorithms learn from thousands of patient records.

Common algorithms include:

- Random Forest
- Support Vector Machine (SVM)
- Logistic Regression
- Decision Tree
- XGBoost
- Deep Neural Networks



Step 4: Prediction

The trained model predicts whether the patient has:

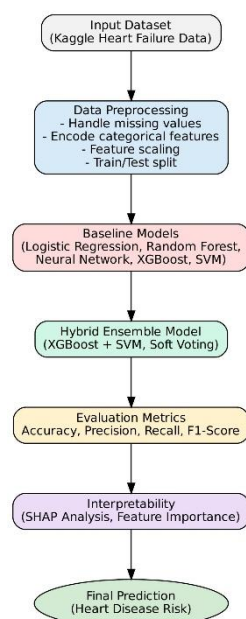
- Low Risk
- Moderate Risk
- High Risk



Step 5: Clinical Decision Support

Doctors review AI predictions together with clinical examinations before making the final diagnosis.

Working Mechanism



8. Data Sources and Processing

Data Sources

- Electronic Health Records (EHR)
- ECG signals
- Blood test reports
- Echocardiography images
- CT scans
- MRI reports
- Wearable health devices
- Smartwatches
- Hospital databases

Data Processing Pipeline

Raw Data



Cleaning



Normalization



Feature Extraction



Model Training



Prediction



Doctor Validation

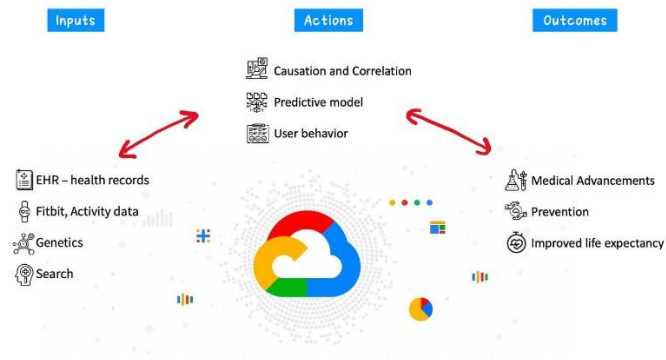
9. Real-World Implementation

Several organizations have successfully implemented AI-assisted heart disease diagnosis.

Organization	AI Application
Mayo Clinic	AI-based cardiovascular disease prediction
Google Health	Deep learning for heart disease detection
IBM	AI-assisted clinical decision support
Philips	AI-enabled cardiac imaging systems
GE HealthCare	AI-powered ECG analysis
Siemens Healthineers	Intelligent diagnostic imaging

These organizations use AI to improve diagnostic speed, reduce errors, and assist cardiologists in making informed clinical decisions.

Real-World Implementation



10. Impact and Benefits

The implementation of AI has significantly improved healthcare.

Clinical Benefits

- Early disease detection
- Faster diagnosis
- Higher prediction accuracy
- Reduced mortality
- Better treatment planning

Hospital Benefits

- Reduced workload
- Improved patient management
- Faster report generation

- Lower healthcare costs
- Better resource utilization

Patient Benefits

- Personalized treatment
- Better monitoring
- Continuous health tracking
- Improved quality of life

11. Challenges and Limitations

Although AI provides many advantages, several challenges remain.

Technical Challenges

- Poor-quality data
- Small datasets
- Model bias
- Overfitting
- High computational requirements

Ethical Challenges

- Patient privacy
- Data security
- AI transparency
- Fairness
- Accountability

Deployment Challenges

- High implementation cost
- Integration with hospital systems
- Regulatory approvals
- Need for skilled professionals
- User acceptance among clinicians

AI should assist doctors rather than replace human expertise.

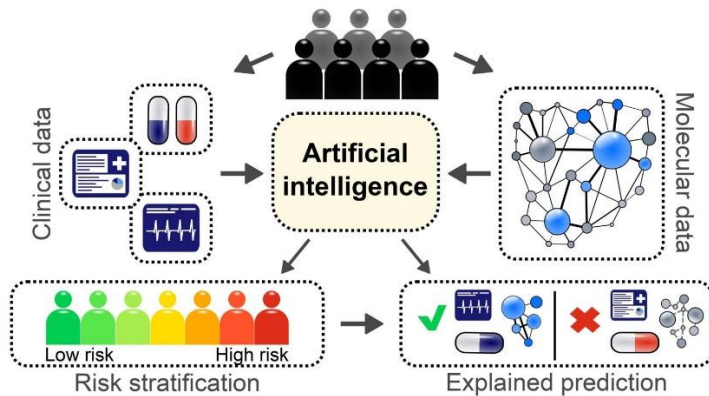
12. Future Scope and Emerging Trends

Future developments include:

- Explainable AI for transparent decision-making
- Federated Learning for privacy-preserving training
- AI integrated with wearable devices
- Real-time patient monitoring
- Personalized precision medicine
- Digital Twin technology for patient simulation
- Generative AI for clinical decision support
- Edge AI for portable healthcare devices
- Multi-modal AI combining medical images, reports, and sensor data
- AI-assisted robotic healthcare systems

Future research aims to create more accurate, secure, and explainable AI systems that improve patient outcomes worldwide.

HEALTH DISEASE PREDICTION



13. Conclusion

AI-assisted medical diagnosis has transformed healthcare by enabling early and accurate detection of heart disease. By combining Machine Learning, Deep Learning, Computer Vision, and Explainable AI, these systems analyze complex medical data efficiently and support clinicians in making informed decisions. Hospitals and healthcare organizations worldwide are adopting AI to improve diagnostic accuracy, reduce treatment delays, and enhance patient care. Although challenges such as privacy, bias, and deployment remain, continuous advancements in AI are expected to make heart disease prediction more reliable, accessible, and effective in the future.

14. References

1. American Heart Association. *Heart Disease and Stroke Statistics Reports*.
2. World Health Organization. *Cardiovascular Diseases (CVDs)*.
3. Esteva, A., et al. (2021). *A Guide to Deep Learning in Healthcare*. Nature Medicine.
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7. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
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